

Initial results v.2.

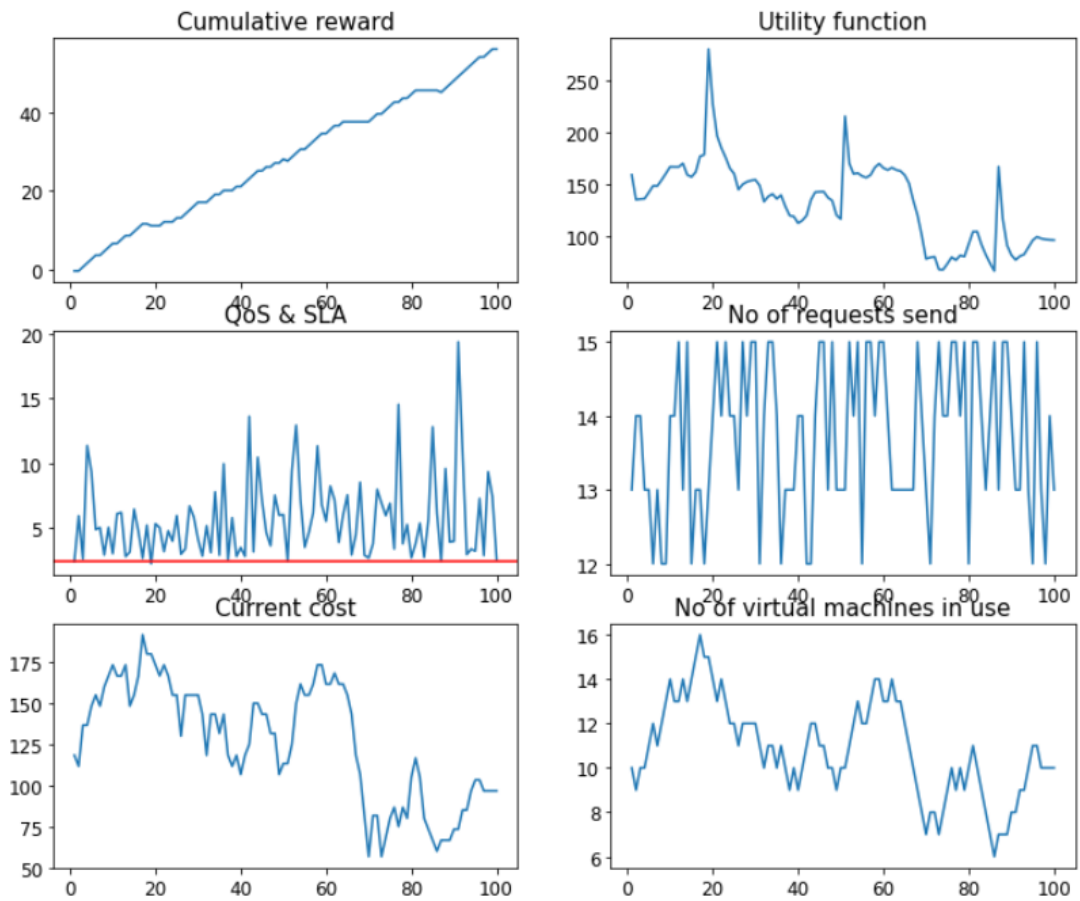
SLA = 2.5, where QoS = requests completed / time (in seconds)

01. Random action agent

Results of testing (1 episode of 100 steps)

Cumulative reward	56
Avg cost per step	126.7
Avg quality of service	5.67
Avg no of requests	13.72
Avg no of virtual machines	10.85

Visualization:

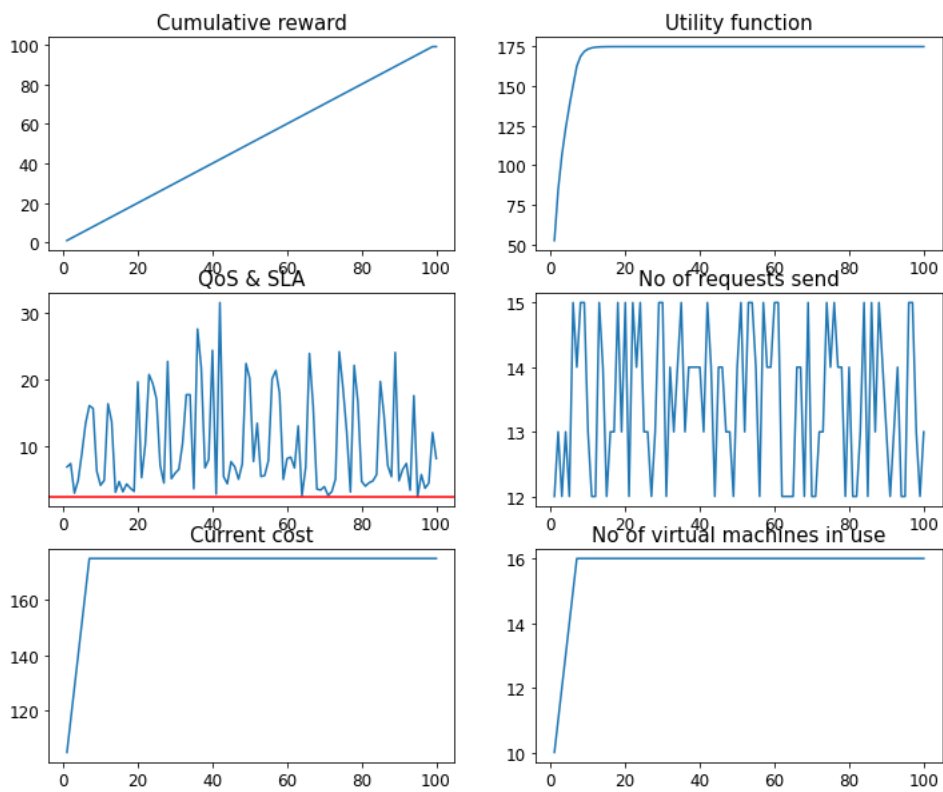


02. Advantage Actor Critic

Results of testing (1 episode of 100 steps)

Cumulative reward	99
Avg cost per step	172.55
Avg quality of service	10.23
Avg no of requests	13.49
Avg no of virtual machines	15.79

Visualization:

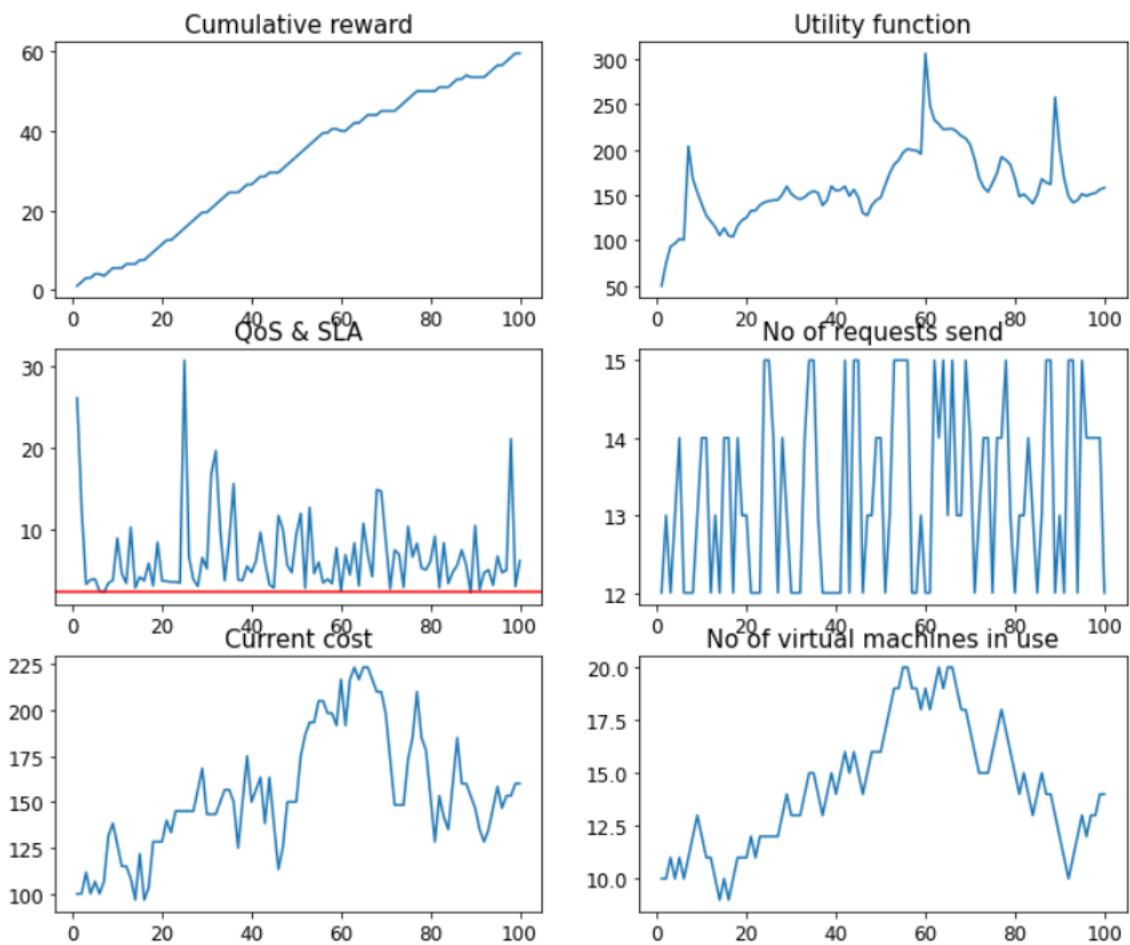


03. Proximal Policy Optimization

Results of testing (1 episode of 100 steps)

Cumulative reward	59.5
Avg cost per step	154.75
Avg quality of service	6.81
Avg no of requests	13.29
Avg no of virtual machines	14.27

Visualization:

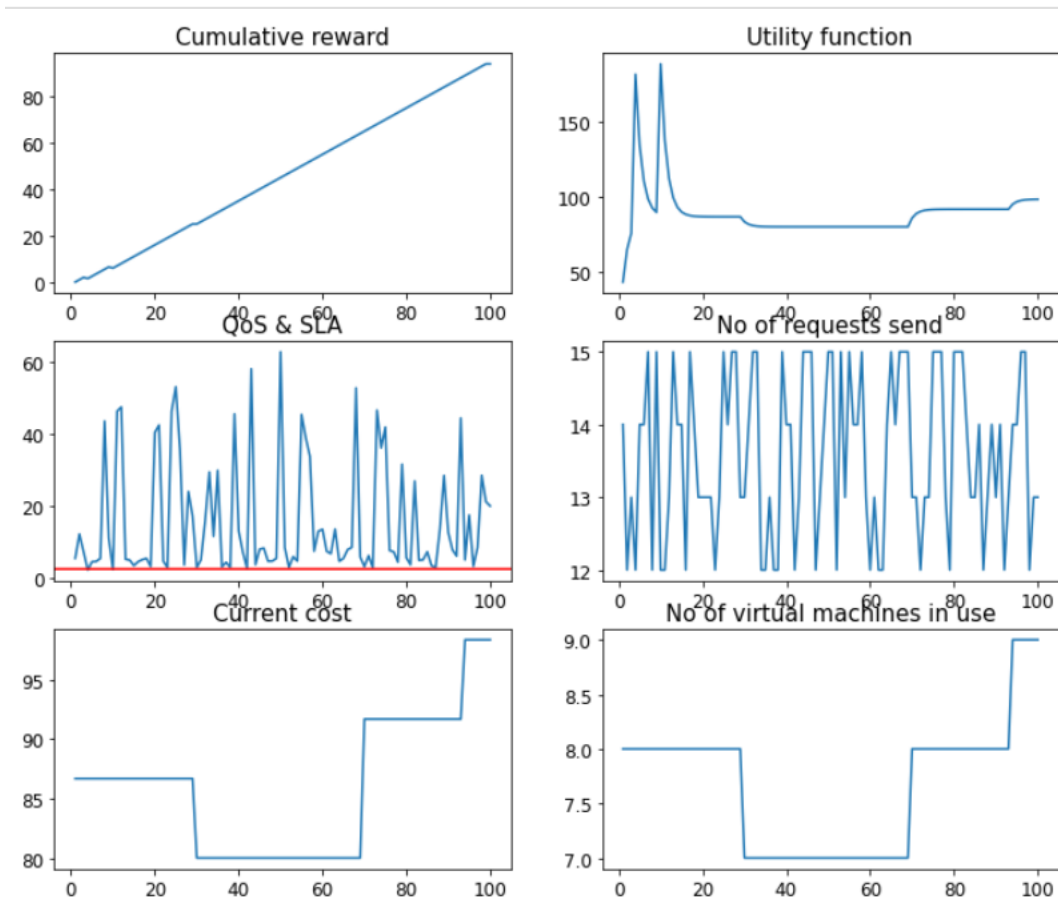


04. Deep Q Network

Results of testing (1 episode of 100 steps)

Cumulative reward	94
Avg cost per step	86.02
Avg quality of service	15.81
Avg no of requests	13.59
Avg no of virtual machines	7.67

Visualization:

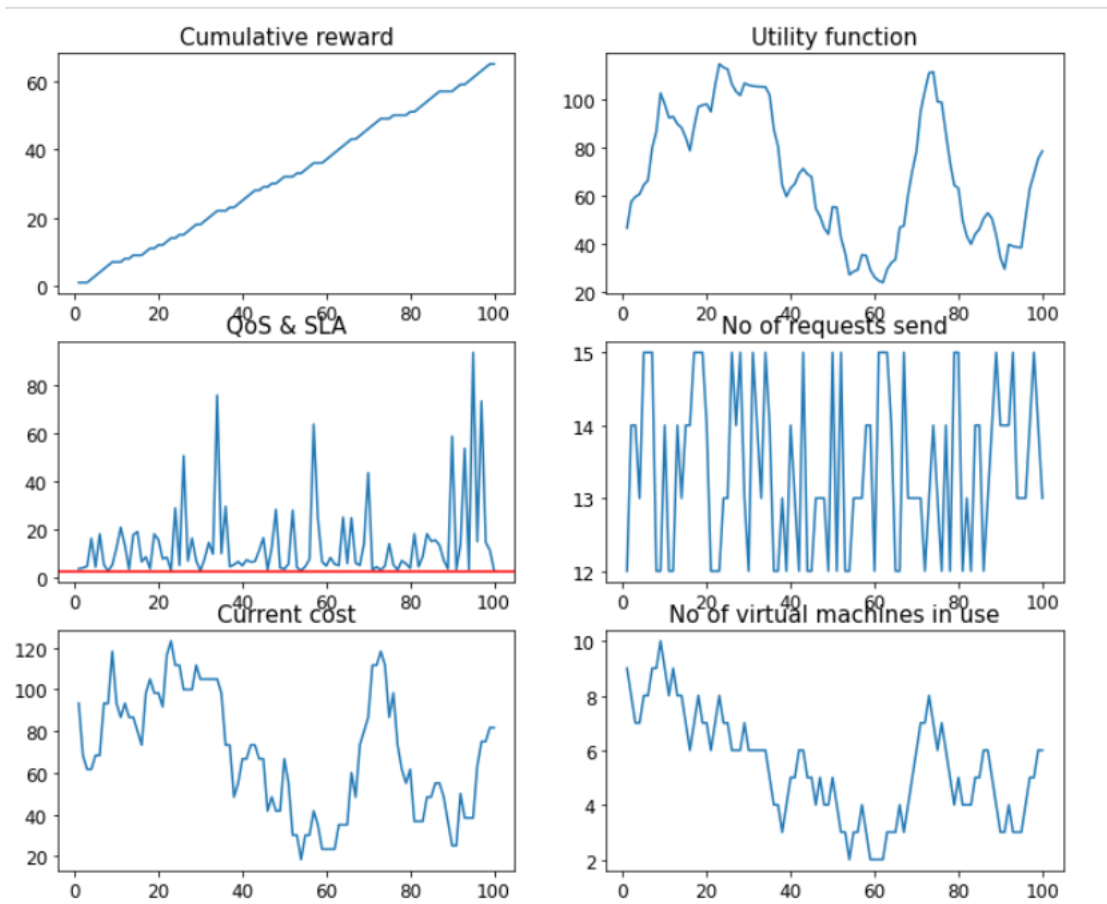


05. Double Deep Q Network

Results of testing (1 episode of 100 steps)

Cumulative reward	65
Avg cost per step	69.2
Avg quality of service	14.03
Avg no of requests	13.4
Avg no of virtual machines	5.37

Visualization:



06. Dueling Deep Q Network

Results of testing (1 episode of 100 steps)

Cumulative reward	59
Avg cost per step	116.33
Avg quality of service	9.23
Avg no of requests	13.25
Avg no of virtual machines	10.5

Visualization:

